

Mie parameters with Feibelman d parameters

We derive the TM scattering coefficients reported in :

Plasmon-emitter interactions at the nanoscale (<https://www.nature.com/articles/s41467-019-13820-z#Sec13>).

We recap the modified boundary conditions:

$$\mathbf{E}_{2,\parallel} - \mathbf{E}_{1,\parallel} = -d_{\perp} \nabla_{\parallel} (E_{2,\perp} - E_{1,\perp})$$

$$\mathbf{H}_{2,\parallel} - \mathbf{H}_{1,\parallel} = i\omega d_{\parallel} (\mathbf{D}_{2,\parallel} - \mathbf{D}_{1,\parallel}) \times \mathbf{e}_n$$

In spherical coordinates, these boundary conditions read (with captial R denoting the radius of the sphere in question):

$$E_{2,\phi} - E_{1,\phi} = -\frac{d_{\perp}}{R \sin(\theta)} \partial_{\phi} (E_{2,r} - E_{1,r})$$

$$E_{2,\theta} - E_{1,\theta} = -\frac{d_{\perp}}{R} \partial_{\theta} (E_{2,r} - E_{1,r})$$

$$H_{2,\phi} - H_{1,\phi} = -i\omega d_{\parallel} (D_{2,\theta} - D_{1,\theta})$$

$$H_{2,\theta} - H_{1,\theta} = i\omega d_{\parallel} (D_{2,\phi} - D_{1,\phi})$$

For the field outside the sphere (where we assume the material is either vacuum or air), we have

$$\mathbf{H}(\mathbf{r}, \omega) = (j_l(r\omega/c) + sh_l^{(1)}(r\omega/c)) \mathbf{L}Y_{lm}(\theta, \phi)$$

For the field inside the sphere, we may only have a spherical Bessel function of the first kind for the radial dependence (since the fields must not diverge at the origin), which means that we have $\mathbf{H}(\mathbf{r}, \omega) = a j_l(nr\omega/c) \mathbf{L}Y_{lm}(\theta, \phi)$

In the following, we take $\mathbf{L} = \mathbf{r} \times \nabla$ (in **Jackson**, for instance, there is an additional factor of $-i$ for the angular momentum operator).

The electric field is given by (noting that $H_r = 0$ everywhere since we are dealing with transverse-magnetic modes):

$$\begin{aligned}\mathbf{E}_{(1,2)} &= \frac{i}{\omega \varepsilon_0 \varepsilon_{1,2}} \nabla \times \mathbf{H}_{1,2} \rightarrow E_{(1,2),\phi} = \frac{i}{\omega \varepsilon_0 \varepsilon_{1,2}} \frac{1}{r} \partial_r (r H_{(1,2),\theta}) \\ E_{(1,2),r} &= \frac{i}{\omega \varepsilon_0 \varepsilon_{1,2}} \frac{1}{r \sin(\theta)} \left(\partial_\theta (\sin(\theta) H_{(1,2),\phi}) - \partial_\phi (H_{(1,2),\theta}) \right) \\ D_{(1,2),\theta} &= -\frac{i}{\omega r} \partial_r (r H_{(1,2),\phi}),\end{aligned}$$

where the numerical subscripts denote whether we are considering the interior or exterior of the scatterer. To evaluate the boundary conditions, we first write the angular momentum operator also in spherical coordinates:

$$\mathbf{L} = \mathbf{r} \times \nabla = e_\phi \partial_\theta - e_\theta \frac{1}{\sin(\theta)} \partial_\phi$$

Which gives us:

$$\begin{aligned}E_r \propto \frac{i}{\omega \varepsilon_0 \varepsilon} \frac{1}{r \sin(\theta)} \left(\partial_\theta (\sin(\theta) \partial_\theta) + \frac{1}{\sin(\theta)} \partial_\phi^2 \right) Y_{lm}(\theta, \phi) = \\ -\frac{i}{\omega \varepsilon_0 \varepsilon r} l(l+1) Y_{lm}(\theta, \phi)\end{aligned}$$

Where by $\propto$ we mean that there's an r dependent combination of spherical Bessel/Hankel functions multiplying the whole thing. More explicitly, we have:

$$\begin{aligned}
 E_{2,r} &= -(j_l(r\omega/c) + sh_l^1(r\omega/c)) \frac{i}{\omega \epsilon_0 r} l(l+1) Y_{lm}(\theta, \phi), \\
 E_{1,r} &= -aj_l(nr\omega/c) \frac{i}{\omega \epsilon_0 \epsilon r} l(l+1) Y_{lm}(\theta, \phi), \\
 E_{2,\phi} &= -\frac{i}{\omega \epsilon_0 r \sin(\theta)} \left[\partial_r(rj_l(r\omega/c)) + s\partial_r(rh_l^{(1)}(r\omega/c)) \right] \partial_\phi Y_{lm}(\theta, \phi), \\
 E_{1,\phi} &= -\frac{i}{\omega \epsilon_0 \epsilon r \sin(\theta)} \left[a\partial_r(rj_l(nr\omega/c)) \right] \partial_\phi Y_{lm}(\theta, \phi), \\
 D_{2,\theta} &= -\frac{i}{\omega r} \left[\partial_r(rj_l(r\omega/c)) + s\partial_r(rh_l^{(1)}(r\omega/c)) \right] \partial_\theta Y_{lm}(\theta, \phi), \\
 D_{1,\theta} &= -\frac{i}{\omega r} a\partial_r(rj_l(nr\omega/c)) \partial_\theta Y_{lm}(\theta, \phi)
 \end{aligned}$$

Therefore, our first boundary condition,

$$E_{2,\phi} - E_{1,\phi} = -\frac{d_\perp}{R \sin(\theta)} \partial_\phi (E_{2,r} - E_{1,r}),$$

gives us (defining $y \equiv r\omega/c$, $x \equiv \sqrt{\epsilon}y$):

$$\begin{aligned}
 \left[\partial_y(yj_l(y) + syh_l^{(1)}(y)) - \frac{a}{\epsilon} \partial_x(xj_l(x)) \right] \left[-\frac{1}{\sin(\theta)} \partial_\phi Y_{lm}(\theta, \phi) \right] = \\
 \frac{d_\perp l(l+1)}{R \sin(\theta)} \left[j_l(y) + sh_l^{(1)}(y) - aj_l(x)/\epsilon \right] \partial_\phi Y_{lm}(\theta, \phi)
 \end{aligned}$$

Simplified to:

$$\partial_y(yj_l(y) + syh_l^{(1)}(y)) - \frac{a}{\epsilon} \partial_x(xj_l(x)) = -\frac{d_\perp l(l+1)}{R} \left(j_l(y) + sh_l^{(1)}(y) - aj_l(x)/\epsilon \right)$$

We introduce the notation, $\bar{d}_\perp = \frac{d_\perp l(l+1)}{R}$ to rewrite the above as:

$$\partial_y(yj_l(y) + syh_l^{(1)}(y)) - \frac{a}{\epsilon} \partial_x(xj_l(x)) = -\bar{d}_\perp \left(j_l(y) + sh_l^{(1)}(y) - aj_l(x)/\epsilon \right)$$

We isolate terms based on whether their coefficients are s (the scattered wave), a (the transmitted wave) or unity (the incident wave) in order to re-arrange the above as:

$$s \left[(yh_l^1(y))' + \bar{d}_\perp h_l^1(y) \right] - \frac{a}{\epsilon} \left[(xj_l(x))' + \bar{d}_\perp j_l(x) \right] = -\bar{d}_\perp j_l(y) - (yj_l(y))'$$

We now use our second boundary condition,

$$H_{2,\phi} - H_{1,\phi} = -i\omega d_{||}(D_{2,\theta} - D_{1,\theta})$$

Which we write as:

$$\left[j_l(y) + s h_l^{(1)}(y) - a j_l(x) \right] \partial_\theta Y_{lm}(\theta, \phi) = - \left(\frac{d_{||}}{R} \right) \left[(y j_l(y))' + s (y h_l^{(1)}(y))' - a (x j_l(x))' \right] \times \partial_\theta Y_{lm}(\theta, \phi)$$

We write this as (introducing the notation $\bar{d}_{||} = d_{||}/R$):

$$s \left[h_l^{(1)}(y) + \bar{d}_{||} (y h_l^{(1)}(y))' \right] - a \left[j_l(x) + \bar{d}_{||} (x j_l(x))' \right] = - \left[j_l(y) + \bar{d}_{||} (y j_l(y))' \right]$$

As a reminder, our first BC is:

$$s \varepsilon \left[(y h_l^{(1)}(y))' + \bar{d}_\perp h_l^{(1)}(y) \right] - a \left[(x j_l(x))' + \bar{d}_\perp j_l(x) \right] = - \varepsilon \left[\bar{d}_\perp j_l(y) + (y j_l(y))' \right]$$

We multiply this equation by $-\left[j_l(x) + \bar{d}_{||} (x j_l(x))' \right]$ and the first equation by $\left[(x j_l(x))' + \bar{d}_\perp j_l(x) \right]$ and add the two equations together, giving us:

$$\begin{aligned} & s \left[\left[h_l^{(1)}(y) + \bar{d}_{||} (y h_l^{(1)}(y))' \right] \left[(x j_l(x))' + \bar{d}_\perp j_l(x) \right] - \right. \\ & \left. \varepsilon \left[j_l(x) + \bar{d}_{||} (x j_l(x))' \right] \left[(y h_l^{(1)}(y))' + \bar{d}_\perp h_l^{(1)}(y) \right] \right] = \\ & \left[\varepsilon \left[j_l(x) + \bar{d}_{||} (x j_l(x))' \right] \left[\bar{d}_\perp j_l(y) + (y j_l(y))' \right] \right. \\ & \left. - \left[(x j_l(x))' + \bar{d}_\perp j_l(x) \right] \left[j_l(y) + \bar{d}_{||} (y j_l(y))' \right] \right] \end{aligned}$$

We write the numerator of s to first order in the d parameters:

$$\varepsilon j_l(x)(y j_l(y))' - j_l(y)(x j_l(x))' + (\varepsilon - 1) \left[\bar{d}_\perp j_l(x) j_l(y) + \bar{d}_\parallel (x j_l(x))' (y j_l(y))' \right]$$

And we expand the denominator of s , also to first order in the d parameters:

$$h_l^{(1)}(y)(x j_l(x))' - \varepsilon j_l(x)(y h_l^{(1)}(y))' - (\varepsilon - 1) \left[\bar{d}_\parallel (x j_l(x))' (y h_l^{(1)}(y))' + \bar{d}_\perp j_l(x) h_l^{(1)}(y) \right]$$

So finally we have:

$$t_{TM} = \frac{\varepsilon j_l(x)(y j_l(y))' - j_l(y)(x j_l(x))' + (\varepsilon - 1) \left[\bar{d}_\perp j_l(x) j_l(y) + \bar{d}_\parallel (x j_l(x))' (y j_l(y))' \right]}{h_l^{(1)}(y)(x j_l(x))' - \varepsilon j_l(x)(y h_l^{(1)}(y))' - (\varepsilon - 1) \left[\bar{d}_\parallel (x j_l(x))' (y h_l^{(1)}(y))' + \bar{d}_\perp j_l(x) h_l^{(1)}(y) \right]}$$

This is consistent with **Equation 4a** in the paper cited in the introduction of this document.

Appendix: A useful relation for the radial component of the electric field

For the TM modes, we know that $\mathbf{H} \propto \mathbf{L} Y_{lm}(\theta, \phi)$

Therefore, the radial component of the electric field looks like:

$$\mathbf{r} \cdot \nabla \times (\mathbf{r} \times \nabla Y_{lm}) = r_i \varepsilon_{ijk} \partial_j \varepsilon_{klm} r_l \partial_m Y = (\varepsilon_{kij} r_i \partial_j) (\varepsilon_{klm} r_l \partial_m) Y = (\mathbf{r} \times \nabla)^2 Y$$